

yüksekliği $\text{fireWidth} \times \text{step}$ Yangın panosu genişliği $\text{fireHeight} \times \text{step}$ Yangın panosu yüksekliği
 $\text{panelWidth} \times \text{step}$ PanelCard genişliği

KOORDİNAT HESAPLARI ————— Bileşen X Y ————— — — F 0 0 (w × h tam
 canvas) H 2 s × 0.3 G 2 headerHeight + 2 + s×0.3 E gridStartX + gridWidth + 4 gridY - s×0.1 R (n) 2
 bscStartY + n × (rackHeight + summaryHeight + s×1.0) HV(n) 2 + n × roomWidth hvacStartY P rooms sonu +
 panelGap hvacStartY FP P sonu + panelWidth + 4 hvacStartY ≡ Bus eaRightX → R.rackXs[0] eaCenterY,
 rackBusY (yatay/çapraz) ► w - s×1.2 s×0.55

DOSYA HARİTASI ————— packages/ui/src/graphics/ system/BESSDiagram/
 BESSDiagram.tsx → Ana birleşik PixiJS canvas BESSDiagram.types.ts → BESSDiagramProps, BESSConfig,
 BSCUnitWithSummary BESSDiagram.utils.ts → calculateBESSConfig(), getBSCLayout(), calcBSCPositions()
 BESSDiagram.drawers.ts → drawGridSymbol, drawBusBar, drawContainerFrame index.ts

elements/ RackCell/ → RackCell.tsx, RackCell.drawers.ts CircuitBreaker/ → CircuitBreaker.tsx,
 CircuitBreaker.drawers.ts DCOutput/ → DCOutput.tsx, DCOutput.drawers.ts RoomCard/ → RoomCard.tsx,
 RoomCard.drawers.ts HvacUnit/ → HvacUnit.tsx, HvacUnit.drawers.ts PanelCard/ → PanelCard.tsx,
 PanelCard.drawers.ts FirePanel/ → FirePanel.tsx, FirePanel.drawers.ts EnergyAnalyzerGraphic/ →
 EnergyAnalyzerGraphic.tsx, EnergyAnalyzerGraphic.drawers.ts Cable/ → Cable.drawers.ts (drawCableBody,
 drawCableArrows)

apps/container-web/src/ pages/ ScadaDashboardPage.tsx → BESSDiagram + LogTerminal kullanan sayfa
 features/ fire-panel/ → useFirePanelData, firePanelApi, types energy-analyzer/ → useEnergyAnalyzerData,
 energyAnalyzerApi, types dashboard/ → useDashboardData (BSC rack verisi) hvac/ → useHvacData,
 hvacUnitsToTmsProps app/ routes.tsx → /scada route tanımı layouts/ SidebarV2.tsx → SCADA sidebar menü
 girişi SidebarV2.types.ts → PageTypeV2 "scada"